KAIST AI4MATH · AI-BASED MATHEMATICAL DISCOVERY · LECTURE 1

Thinking about Thinking

Humans, Machines, Mathematics

Sang-hyun Kim · KIAS School of Mathematics · KAIST AI4Math

What is thinking?

1958 · THE FEELING OF POWER

Forgetting Multiplication

```
  17
× 23
----
  51
 34
----
 391
```

Far future: all calculation by computer · multiplication forgotten

Aub, a low-grade technician: ‘rediscovers’ 17 × 23 = 391 with pencil and paper

Objection: calculation is, by definition, what computers do

— Isaac Asimov, “The Feeling of Power” (1958) · paraphrased

Can we entrust thinking to machines?

Can machines think?

Can machines do mathematics?

I. Dream of a Thinking Machine

1679 · HANOVER

Reasoning as Calculation



G. W. Leibniz
(1646-1716)

An instrument to strengthen the mind more than lenses strengthen the eye

— Leibniz, c. 1679 (paraphrased)

AI as a revolutionary lens — as telescopes and microscopes extended perception

— Pushmeet Kohli (Google DeepMind), 2026 · paraphrased



Leibniz's Stepped Reckoner (1673 – 1694)

First design for a four-operation calculator · hand crank

Multiplication = repeated addition · division = repeated subtraction

1900 · 1928 · ENTSCHEIDUNGSPROBLEM

Hilbert's Decision Problem



David Hilbert (1862-1943)

A mechanical procedure to decide every mathematical statement?

= An algorithm that solves every mathematical problem?

Where are the limits of machines?

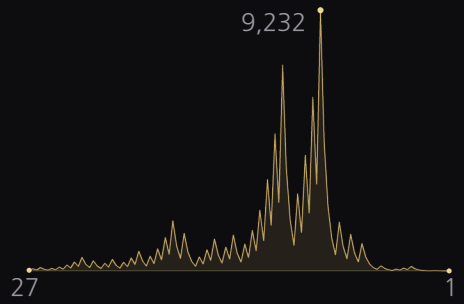
1937 · COLLATZ

The 3n + 1 Conjecture

Even: n → n/2 · Odd: n → 3n + 1

6 → 3 → 10 → 5 → 16 → 8 → 4 → 2 → 1

27 → 82 → 41 → 124 → ⋯ → 9,232 → ⋯ → 1 (111 steps)



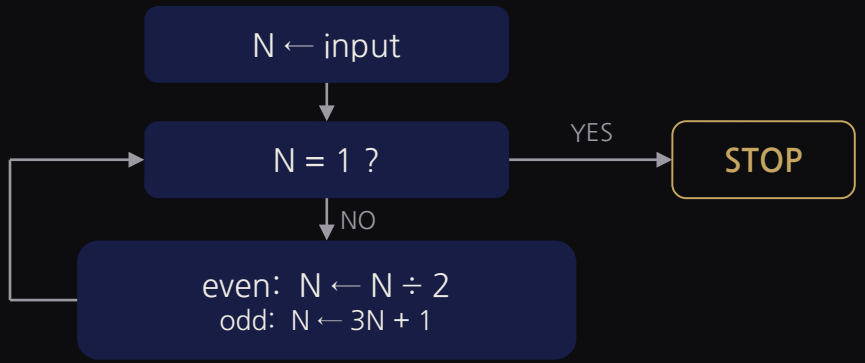
Conjecture: every positive integer reaches 1

Open since 1937 · verified up to 2,361,183,241,434,822,606,848

Verification ≠ proof

"Mathematics is not yet ready for such problems." — Paul Erdős

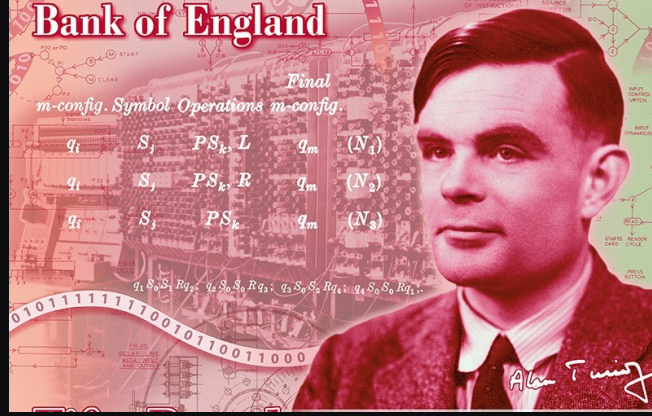
The 3n + 1 Program



N = 6 → 8 steps, halts ○
N = 7 → 16 steps ○
N = 27 → 111 steps ○
N = 97 → 118 steps ○
…
All up to 2.36×10^{21} ○
The next one ?

Deciding ‘always halts’ = solving the 3n + 1 conjecture

Undecidability of Halting



Alan Turing (1912-1954)
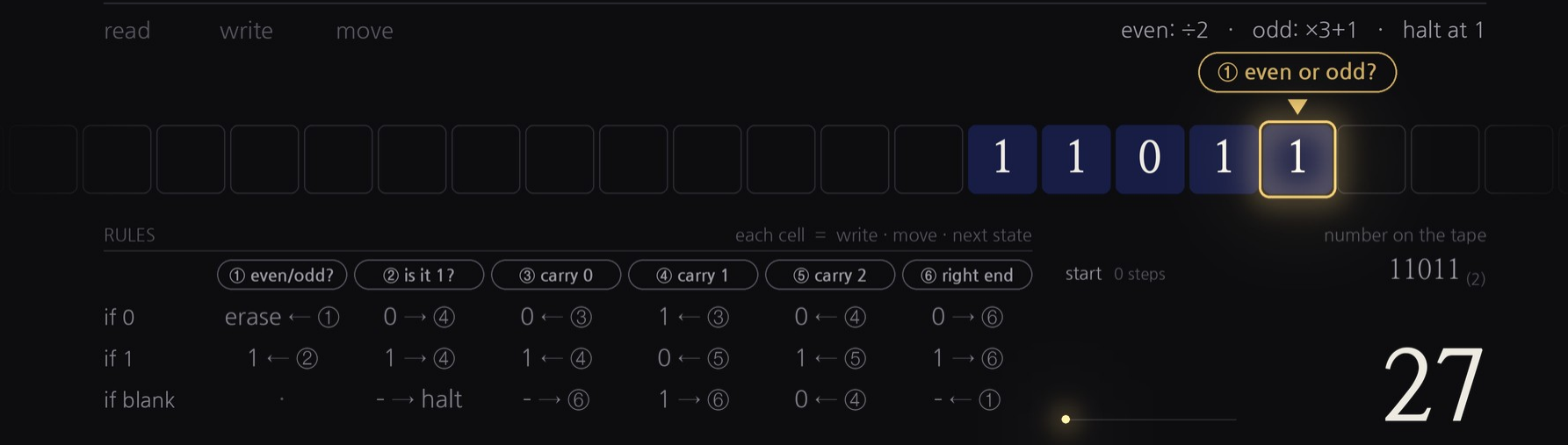
Defining ‘mechanical computation’ — Turing, age 23

Turing machine: the theoretical model of modern computers

A general algorithm to decide halting — does not exist

Some problems no computer can solve

The Turing Machine



Read · write · move one cell — is this ‘thinking’?

SEP 1930 · KÖNIGSBERG

The Incompleteness Theorem



Kurt Gödel
(1906-1978)

"We must know. We will know." — Hilbert, Sep 8, 1930

The day before, same city — Gödel (24) announces incompleteness

Consistent formal arithmetic: undecidable statements exist

Formal proof alone cannot reach all mathematical truth

Either mathematics exceeds the mind, or the mind exceeds any machine

— Gödel, Gibbs Lecture, 1951 (paraphrased)

I. Dream of a Thinking Machine

II. From Dream to Machine

Can machines

go beyond

mechanical thinking?

Intelligent Machinery, A Heretical Theory*

A. M. TURING

'You cannot make a machine to think for you.' This is a commonplace that is usually accepted without question. It will be the purpose of this paper to question it.

Commonplace: no machine can think for us

The paper's aim: to question it

— Alan Turing (1951)

Deep learning · Reinforcement learning

Linear algebra: data transformation

Calculus: gradient descent

Probability: uncertainty



All undergraduate math —
yet why does it work?

2016 · ALPHAGO MOMENT

The AlphaGo Moment



Four Seasons Hotel Seoul

March 2016

200 million	viewers worldwide
4-1	AlphaGo wins (vs. Lee Sedol, 9-dan)
	“…my entire world was collapsing.” — Lee Sedol, 2024
Move 78	Game 4 · AlphaGo’s estimated probability: 1/10,000

LLMs: The Geometry of Relations

6. Second-order word approximation. The word transition probabilities are correct but no further structure is included.

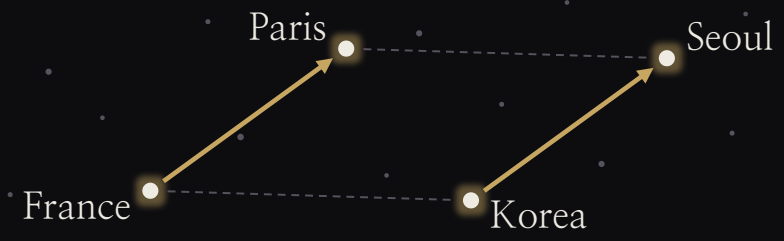
THE HEAD AND IN FRONTAL ATTACK ON AN ENGLISH WRITER THAT THE CHARACTER OF THIS POINT IS THEREFORE ANOTHER METHOD FOR THE LETTERS THAT THE TIME OF WHO EVER TOLD THE PROBLEM FOR AN UNEXPECTED.

Shannon (1948) — second-order word approximation: next word depends only on the previous word

LLM: next-word prediction

"I wrote the **date**, ate a **date**, on a **date**."

$$\text{Paris} - \text{France} \approx \text{Seoul} - \text{Korea}$$



1948 · BELL LABS · SHANNON

A ‘Language Model’ from Book and Pencil

Procedure: a word on a random page → find it on another page, take the next word → repeat

THE HEAD AND IN FRONTAL ATTACK ON AN ENGLISH WRITER THAT THE CHARACTER OF THIS POINT IS THEREFORE ANOTHER METHOD …

Next stage: interesting, but the labor becomes enormous — Shannon, 1948 (paraphrased)

Same year, same lab: the transistor unveiled

II. From Dream to Machine

III. Mathematics by Machine

2024 → 2025 · INTERNATIONAL MATHEMATICAL OLYMPIAD

IMO: From Silver to Gold

Silver

IMO 2024 — Silver

AlphaProof + AlphaGeometry 2
4/6 problems (28 points)

Manual Lean formalization

Up to 3 days

Gold

IMO 2025 — Gold

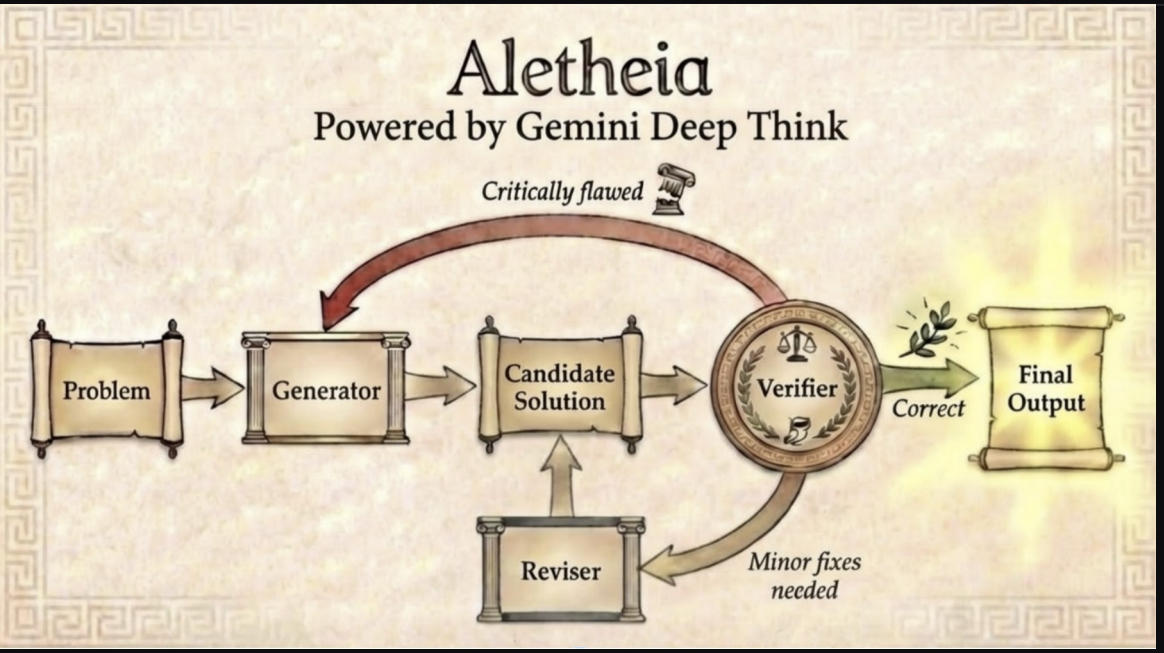
Gemini Deep Think
5/6 problems (35 points)

Natural-language proofs

4.5 hours (same as contestants)

2026 · ALETHEIA — ἀλήθεια, TRUTH

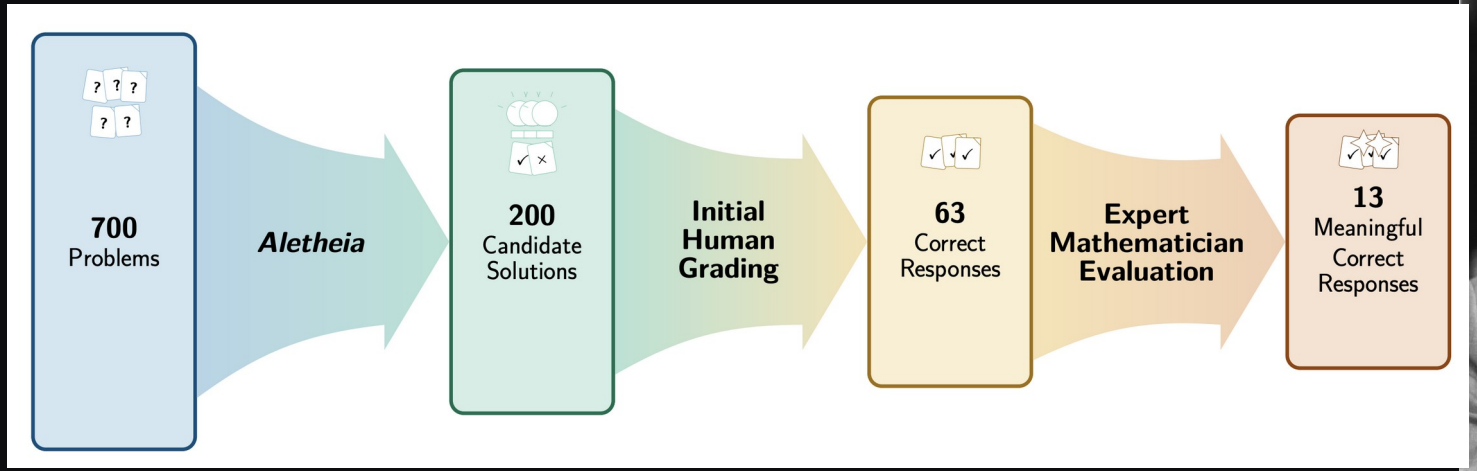
Aletheia: A Math Research Agent (Google)

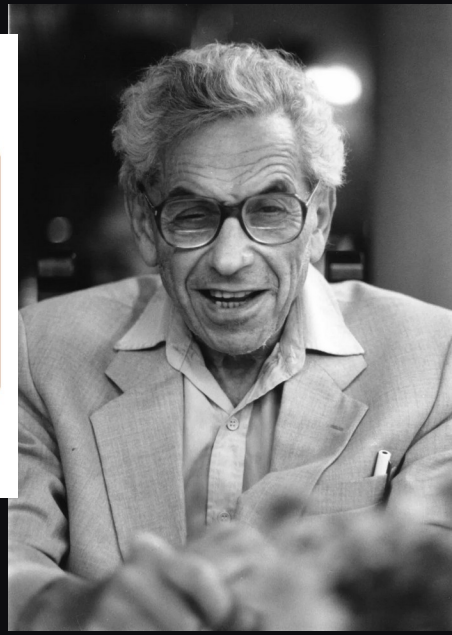


Gemini-based research agent

Generator → Verifier → Reviser

Mathematicians + engineers





Paul Erdős (1913-1996)

Erdős	~1,500 papers · ~1,000 open problems
Aletheia	700 problems → 63 correct → 13 meaningful → 4 new solutions
Problem **1051**	AI (2 pp.) → human (5 pp.) → AI (5 pp.) → human (20 pp.)
	One small problem → a deep theorem (LMS journal)

FEB 6-13, 2026 · FIRSTPROOF

FirstProof Challenge

Mathematicians' unpublished research problems

10 problems · 7 days — Google, OpenAI: 6 each

Aletheia: Problem 7 solved (13-year-old conjecture)

- Problem author: fully correct
- Results used:

  Selberg, Cartan, Leray, Borel, Atiyah, Novikov, Kasparov…

- Highly compressed combination of theorems and theories

  Reads like a top expert's effortless solution

Taking the inverse limit as $N \to \infty$ (over the parity-appropriate sequence of spheres), the true equivariant pullback $f^*_{\mathbb{Z}/2} : H^*_{\mathbb{Z}/2}(X_0; \mathbb{F}_2) \to H^*_{\mathbb{Z}/2}(M_0; \mathbb{F}_2)$ is forced to be injective. By the exactness of localization, passing to the localized rings by inverting the polynomial generator $u \in H^1(B\mathbb{Z}/2; \mathbb{F}_2)$ preserves this injectivity:

$$S^{-1} f^*_{\mathbb{Z}/2} : S^{-1} H^*_{\mathbb{Z}/2}(X_0; \mathbb{F}_2) \hookrightarrow S^{-1} H^*_{\mathbb{Z}/2}(M_0; \mathbb{F}_2).$$

By the Atiyah–Borel Localization Theorem, the localized equivariant cohomology of a space is isomorphic to the localized equivariant cohomology of its fixed-point set:

$$S^{-1} H^*_{\mathbb{Z}/2}(Y; \mathbb{F}_2) \cong H^*(Y^{\tilde{\gamma}}; \mathbb{F}_2) \otimes_{\mathbb{F}_2} \mathbb{F}_2[u, u^{-1}].$$

Because $X_0^{\tilde{\gamma}} \neq \emptyset$, the localized domain evaluates to a non-zero module. Because $M_0^{\tilde{\gamma}} = \emptyset$, the localized target is identically zero. An exact injection from a non-zero module into a zero module is impossible. Therefore, D must rigidly be an **even integer**.

Step 3: Higher Signature Rigidity via the Strong Novikov Conjecture
We evaluate the symmetric higher signature of M_0 within the real C^*-algebra K-theory $KO_d(C^*_r(\Gamma_0; \mathbb{R}))$ to establish a fundamentally conflicting constraint.
Because $\tilde{M}$ and X are $\mathbb{Q}$-acyclic, the Universal Coefficient Theorem guarantees they are $\mathbb{R}$-acyclic. Thus, their real cellular chain complexes $C_*(\tilde{M}; \mathbb{R})$ and $C_*(X; \mathbb{R})$ constitute finite-length projective free resolutions of the trivial module $\mathbb{R}$ over the real group ring $\mathbb{R}\Gamma_0$. By the Fundamental Lemma of Homological Algebra, the chain map $C_*(\tilde{f}; \mathbb{R})$ is inherently a strict chain homotopy equivalence over $\mathbb{R}\Gamma_0$.

Google Aletheia output

1946 → MAY 2026 · UNIT DISTANCE PROBLEM

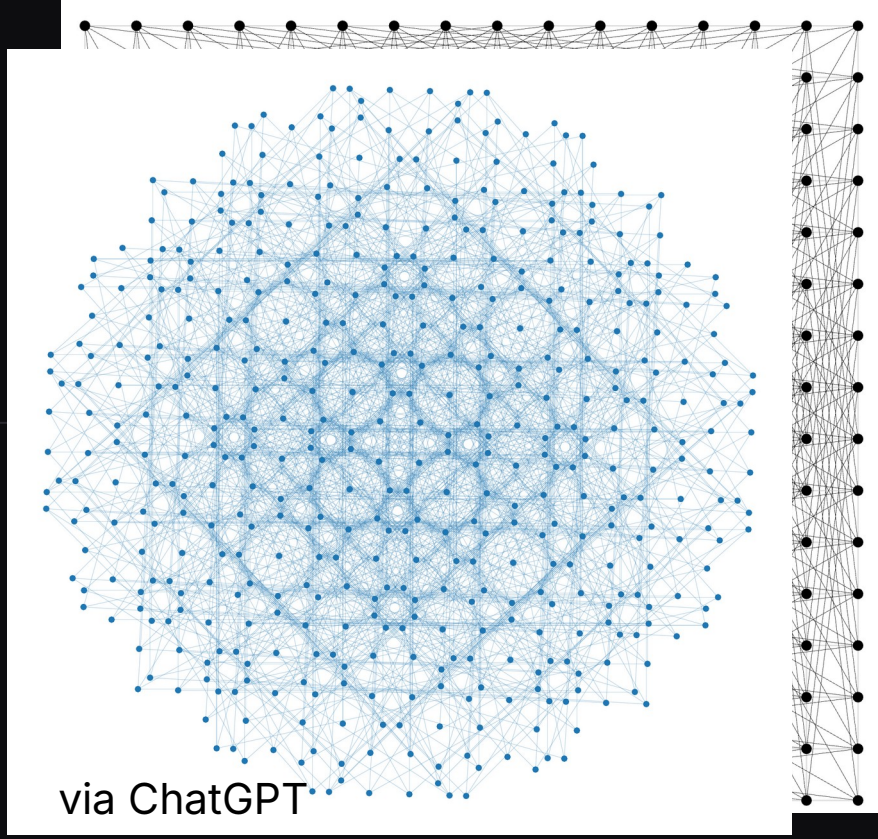
Unit Distances (OpenAI)

Erdős (1946)

n points in the plane: max number of pairs at distance 1

80-year belief: lattice configurations essentially optimal

- Disproved by AI — humans supplied only the problem
- Geometry × number theory — combined autonomously
- A milestone for AI in mathematics — Tim Gowers (Cambridge)



MAY 2026

Convexity Problem

Michel Talagrand (Abel Prize 2024)

KIAS lecture notes · “My favorite problems” (2025)

Half: Antoine Song (Caltech), Feb 2026

The rest: grad student Dongming Hua — “Please continue.”

Please continue.

Whose discovery is this?

can we
talagrand's

That is the exact point where a genuine “subgaussian MSS” theorem must enter. The Bessel-block version is provable now; the weighted ray/exposure upgrade is the missing step that would turn Song’s framework into a solution of Talagrand’s convexity problem.

출처

k in
k for newer

please try to prove the weighted ray upgrade.

I’ll focus directly on the weighted ray upgrade, recasting it as a finite-dimensional Gram/Loewner domination problem. This should reveal whether minimax averaging can yield one labeled Gaussian partition with a contraction.

2.14; that

My favorite problems

Michel Talagrand[1]

Abstract

I discuss some problems I studied but could not solve.

JUL → OCT 2026

July – October 2026

Jul 19 — Jacobian disproved
Anthropic

Aug 1 — Ten advances
OpenAI

Aug 12 — Hadamard 668
Anthropic

Sep 8 — Navier-Stokes
OpenAI

Oct 6 — 372 problems
OpenAI

3 years ago — unreliable at adding two numbers

1 year ago — IMO gold

Now — resolving major open problems on its own

— Daniel Litt (Toronto), Sep 2026 · paraphrased

Jacobian Counterexample (Anthropic)

"hello there the jacobian conjecture is false thanx to my close friend akhil for asking about it and my other close friend fable for working during the world cup final"

— Levent Alpöge (Anthropic), Jul 19, 2026 · 'fable' = AI model Claude

Conjecture (Keller, 1939) Is every polynomial map with nonzero constant Jacobian determinant invertible?
Counterexample: 3 variables, degree 7, 3-to-1

$$F(x,y,z) = \Big((1+xy)^3 z + y^2(1+xy)(4+3xy),\; y + 3x(1+xy)^2 z + 3xy^2(4+3xy),\; 2x - 3x^2 y - x^3 z \Big)$$

87 years to find · minutes to check

2 variables: still open

$$\det DF = -2$$

$$F\left(0,\,0,\,-\tfrac{1}{4}\right) = F\left(1,\,-\tfrac{3}{2},\,\tfrac{13}{2}\right) = F\left(-1,\,\tfrac{3}{2},\,\tfrac{13}{2}\right) = \left(-\tfrac{1}{4},\,0,\,0\right)$$

Ten Advances (OpenAI)

① **High-dimensional sphere packing**
Density upper bound

② **Binary and spherical codes**
Error-correction limits

③ **Non-sofic groups**
Existence proved

④ **Connes rigidity conjecture**
Disproved

⑤ **Arithmetic circuit complexity**
New lower bound

⑥ **Quantum parallel repetition**
Two-player games

⑦ **Closest vector problem**
Lattice cryptography

⑧ **Ehrhart volume conjecture**
Convex-body volume

⑨ **Multicolor Ramsey numbers**
Erdős #183

⑩ **Extremal function conjecture**
Erdős #146 · #180

Unreleased model ‘Astra’: proof search → human-model co-writing → Lean verification

Compute cost: ~$2,000 for 10 problems

Open Problems

3n + 1 conjecture	Posed 1937 · verified to 2.36×10^{21} · no proof
Jacobian conjecture	3-variable counterexample (2026) · 2 variables open
Aletheia's 700 problems	4 new solutions · most still open
FirstProof: 4 unsolved	AI reported 'not solved' instead of inventing answers

Most problems remain unsolved

SEP 8, 2026 · NAVIER-STOKES

AI's First Millennium Problem (OpenAI)

Navier-Stokes equations (1822-): fluid motion · Millennium Problem

Question: finite-time singularity (blow-up) from smooth initial data?

OpenAI: singularity under smooth external force — parts (C)·(D) of the official problem

~10,000 AI agents · 88 hours · Lean verification in 17 hours

Key strategy: Córdoba · Martínez-Zoroa · priority dispute · verification ongoing

An answer — but understanding?

OCT 6, 2026 · OPENAI / MATH

372 Problems · 722 Papers (OpenAI)

Unreleased internal model · ~4,000 problems → results on 372 (bulk release on GitHub)

Full and partial results · ~3 hours of ChatGPT Pro compute per result on average

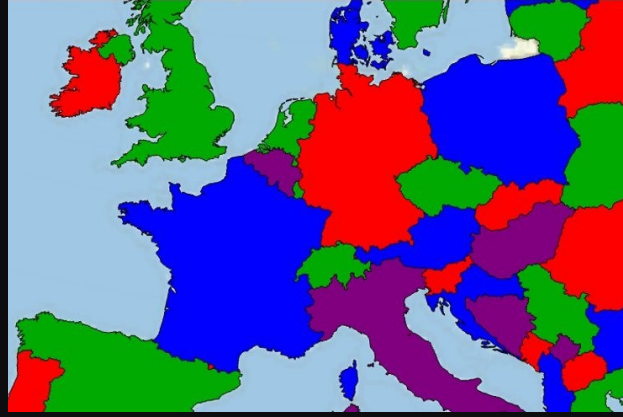
E.g. unique games conjecture · Mahler · free group factors · Kaplansky direct finiteness (claims)

Not yet peer-reviewed · partly Lean-verified · 3 papers withdrawn the next day (sign error)

722 papers — who will read them?

1852 → 1976 · FOUR COLOR THEOREM

The Four Color Theorem (1976)



Planar maps, adjacent regions in different colors — are four colors enough?

Open for 124 years → 1976: 1,936 configurations checked by computer

First major theorem proved by computer

Reaction: unease and debate

If no one on Earth fully understands a proof — is it a proof?

1994 · ON PROOF AND PROGRESS IN MATHEMATICS

The Discovery – Understanding Gap

The Four Color debate: not doubt about its truth — but the wish for human understanding of the proof

Computing 10,000 primes: the goal was never a pile of answers — it was understanding

— William Thurston (Fields Medal 1982), "On Proof and Progress in Mathematics" (1994) · paraphrased

Answers pile up — who does the understanding?

III. Mathematics by Machine

IV. The Thinking Human

JUN → SEP 2026 · THE COMMUNITY RESPONDS

Statements by Mathematicians

Jun 2 Leiden Declaration on AI and Mathematics — 4,300+ signatories, endorsed by IMU

Disclose AI use · cite sources · humans accountable · peer review, not press releases

Sep 11 28 Fields Medalists: “A Severe Misalignment of AI in Mathematics”

Concern: true/false verdicts, mass-produced ever faster, eroding the ground where new ideas grow

Urgent for society as a whole — other intellectual work is next

— Fields Medalists’ statement, Sep 11, 2026 · paraphrased

Can machines think?

— Alan Turing, “Computing Machinery and Intelligence” (1950)

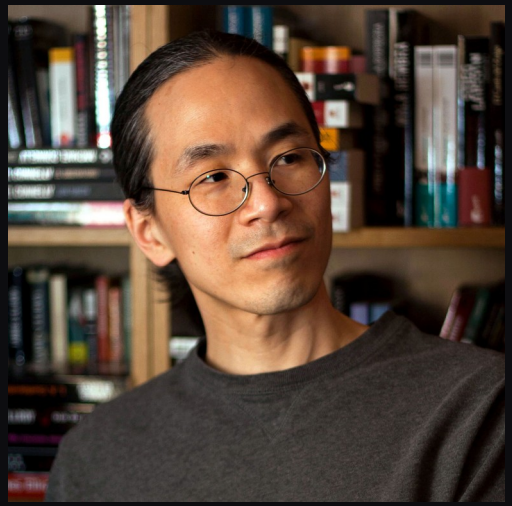
Are machines thinking?

— cf. Ludwig Wittgenstein, Remarks on the Foundations of Mathematics (1956)

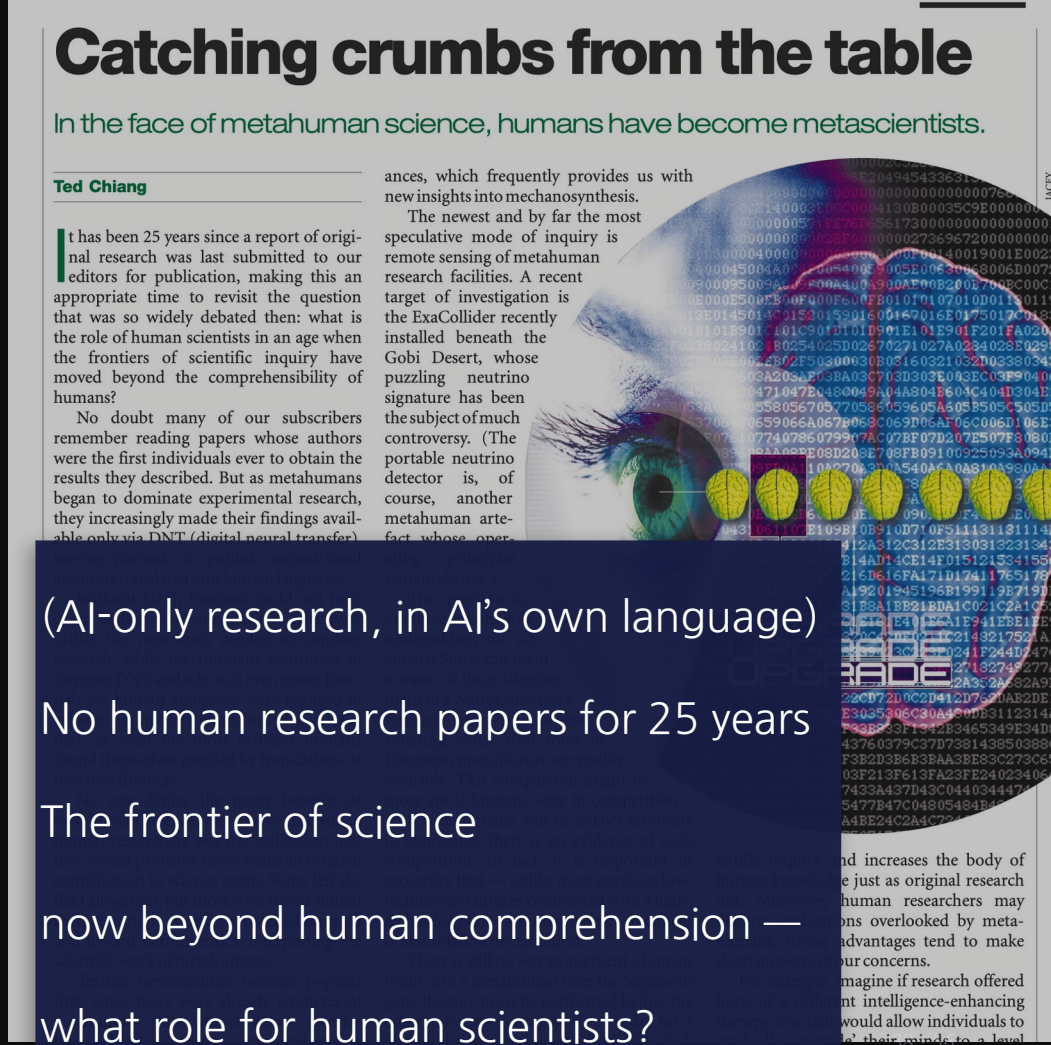
Should we think?

Catching Crumbs from the Table

Ted Chiang, Nature (2000)



Ted Chiang (b. 1967)



Catching crumbs from the table

In the face of metahuman science, humans have become metascientists.

Ted Chiang

It has been 25 years since a report of original research was last submitted to our editors for publication, making this an appropriate time to revisit the question that was so widely debated then: what is the role of human scientists in an age when the frontiers of scientific inquiry have moved beyond the comprehensibility of humans?

No doubt many of our subscribers remember reading papers whose authors were the first individuals ever to obtain the results they described. But as metahumans began to dominate experimental research, they increasingly made their findings available only via DNT (digital neural transfer)

ances, which frequently provides us with new insights into mechanosynthesis.

The newest and by far the most speculative mode of inquiry is remote sensing of metahuman research facilities. A recent target of investigation is the ExaCollider recently installed beneath the Gobi Desert, whose puzzling neutrino signature has been the subject of much controversy. (The portable neutrino detector is, of course, another metahuman artefact whose oper-

(AI-only research, in AI's own language)

No human research papers for 25 years

The frontier of science

now beyond human comprehension —

what role for human scientists?

2000 · NATURE — CATCHING CRUMBS FROM THE TABLE

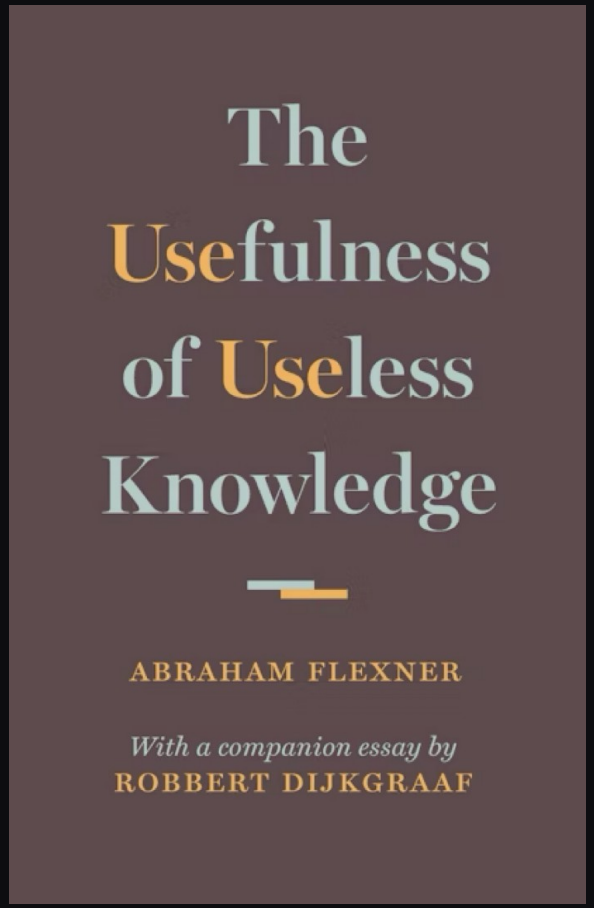
Metahuman science: no cause for intimidation.

The technology behind metahumans

was a human invention —

by people no cleverer than us.

— Ted Chiang, “Catching Crumbs from the Table,” closing lines · paraphrased



Throughout the history of science,

the great discoveries that ultimately

served humanity

came mostly from people moved

not by a wish to be useful,

but simply by curiosity.

(Abraham Flexner, 1939 · paraphrased)

Please continue.

So, what should

thinking be?

AIM60003 · AI-BASED MATHEMATICAL DISCOVERY · FALL 2026

Doing Mathematics with AI

Syllabus · Discussion · Visualization Project · Math Project

Sang-hyun Kim · KIAS School of Mathematics · KAIST AI4Math

SYLLABUS · TUESDAYS 10:30-12:30 · IN ENGLISH

Four Lectures

1	**Oct 13**	Instructor's lecture	Thinking about Thinking · two visualizations · Thompson's group F
—	**Mid term**	No class	
2	**Oct 27 ??**	Class discussion	Five topics on AI and the future of mathematics · five groups of five
3	**Nov 3**	Visual / Math projects I	Visualization (5 min / each) and math (5 min / each) projects
4	**Nov 17 ??**	Visual / Math projects II	Visualization (5 min / each) and math (5 min / each) projects

Lecture 2, 4: Oct 29 – 30 (Th/Fr), Nov 3 (Tue): 9am,
Nov 16 (M), 18 – 20 (WThF), 23 (M), 30 – Dec 4 (M – F), 7, 9, 10 (MWTh)

Assessment and Deadlines

COMPONENTS

Discussion presentation	Group of 5 · Lecture 2	25%
Visualization project	Pair · Lecture 3 or 4	30%
Math project	Pair · Lecture 3 or 4	30%
Participation	Individual · Lectures 2–4	15%

25 students: one three-member team for each project type · same or different partners for the two projects · submissions via KLMS

Agenda

1	**Class dates**	Lecture 2 and Lecture 3
2	**Five groups of five: discussion topics**	Email me three choices by Oct 20
3	**Discussion slides**	Email me by 1 day before Lecture 2
4	**Visualization OR Math Research**	Email me: **proposals** by Oct 20 **slides** by 1 day before presentation **notes** (1-3 pages) after the presentation

LECTURE 2 · OCT 27 · 20 MIN PER TOPIC: 10 PRESENTATION + 10 DISCUSSION

Five Discussion Topics

10:35	1	**AI and Mathematics as a Scientific Revolution**	A revolution? Which paradigms — and what is lost?
10:55	2	**The Qualities of Future Mathematicians**	Which qualities gain value? What survives of what we love?
11:15	3	**The Future of the Mathematical Community**	Who leads, who gets credit, who controls compute?
11:35	4	**The Art of New Mathematics**	Which skills matter when AI searches, computes, proves?
11:55	5	**The Ethics of AI-Assisted Research**	Who is an author? Who answers for errors?

LECTURE 2 · GROUPS OF FIVE · 5 MIN · UNDER 10 SLIDES

Discussion: Five Short Talks

What are facts and terms?

Why does it matter now?

Traditional position?

Oppositional position?

Seed questions, open questions, decision-now questions?

Prediction and value?

A fact, a prediction, or a value?

Reading List

YEAR	READING	WHY READ IT NOW	TOPICS
1936	**Turing**, "On Computable Numbers"	Computability and undecidability	1, 4
c. 1951	**Turing**, "Intelligent Machinery, A Heretical Theory"	Learning machines may outstrip us	1, 2
1951	**Gödel**, Gibbs Lecture: "Some Basic Theorems …"	Gödel's dilemma on mind and machine	2, 4
1956	**Wittgenstein**, *Remarks on the Foundations of Mathematics*	Surveyability; proof vs. calculation	1, 4
1962	**Kuhn**, *The Structure of Scientific Revolutions*	Paradigms and what revolutions lose	1
1992	**Smullyan**, *Gödel's Incompleteness Theorems*	Incompleteness, rigorous and compact	1
1993	**Jaffe and Quinn**, "Theoretical Mathematics"	Labeling speculation; rules for credit	3, 5
1994	**Thurston**, "On Proof and Progress in Mathematics"	Mathematics as human understanding	2, 3, 4
2017	**Stewart**, "Counting infinity", *Infinity: A Very Short Introduction*	Cantor: opposed at first, then loved	1, 2
2019	**Sutton**, "The Bitter Lesson"	Scaling beats built-in human knowledge	1, 3, 4

Topics: the five discussion topics of Lecture 2 · full references in the syllabus

Mathematical Aesthetics

YEAR	READING	WHY READ IT NOW
1908	**Poincaré**, “Mathematical Creation”, in *Science and Method*	The aesthetic sense as the sieve of discovery
1940	**Hardy**, *A Mathematician’s Apology*	Beauty, seriousness and permanence
1947	**von Neumann**, “The Mathematician”	A driving force, and the risk of drifting from empirical sources
1976	**Lakatos**, *Proofs and Refutations*	How definitions and rigor evolve through counterexamples
1997	**Rota**, “The Phenomenology of Mathematical Beauty”	Beauty as the experience of enlightenment

Visualization Project

Aim	A piece of mathematics made visible: a definition, a theorem, an example or a process
Audience	Chosen by the pair and stated in the proposal
Medium	A self-contained HTML page, no installation · a notebook or video when better suited
Examples	Fourier and Diffusion
Submissions	Proposal Oct 25 · link before the lecture · one-page note + prompt log one week after
Evaluation	Correctness · clarity for the audience · design and craft · honest account of AI's part

What can it show that a proof or a static figure cannot?

Visualization: Note and Demo

ONE-PAGE NOTE

1 Goal — what someone should see or understand

2 Mathematical content — what it carries and omits

3 Target audience — who, and what they already know

4 Prompts and steps — order, failures, fixes

5 Food for thought — AI's part, human judgment

FOUR-MINUTE DEMO

Q1 To whom is it addressed?

Q2 What should be delivered?

Q3 Why a visualization — and not a proof or a figure?

Q4 What are the aesthetic criteria?

Q5 What was the main technical difficulty, and the way past it?

Math Project Candidates

TYPE	SUCCESS LOOKS LIKE	MAIN RISK
Open problem from a list	Proof, counterexample, or sharper partial result	Already solved · false novelty
Question from a recent paper	An answer, a special case, or a reduction	Misreading the question
Own research problem	A lemma or step toward the thesis problem	Confidentiality
Known theorem, new route	A new route, compared with the literature	Known proof, unannounced
Computation or formalization	Data, conjecture, counterexample, Lean proof	Output without understanding

Type chosen in the proposal (Oct 25) · fallback within the same problem

MATH PROJECT NOTE: 1 PAGE + SLIDES

Design Questionnaire

Goal description	Solve this? Simplify this? Make a breakthrough toward this?
Initial prompt	Start research?
Continuation prompt	Please continue?
Staggering alert / reaction	If no advances for five times?
Context overflow / reaction	Write a hand-off for the next chat window?
Verification criteria	If the solution passes three independent agent reviewers' checks?
Check points	Report minor / major progresses?
Literature / human inputs	Thoroughly search the literature for similar results?

MATH PROJECT NOTE: 1 PAGE + SLIDES · EXAMPLE

Design Questionnaire

GOAL DESCRIPTION

Prove $PL_+(S^1)$ contains $\mathbb{Z}^2 * \mathbb{Z}$ by this construction

$\Rightarrow$ Prove $PL_+(S^1)$ does not contain $\mathbb{Z}^2 * \mathbb{Z}$

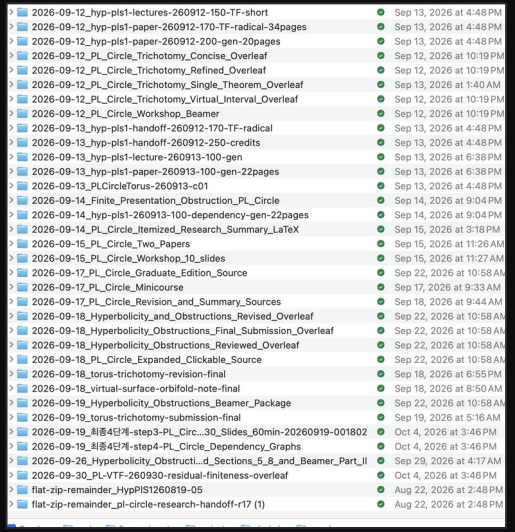
$\Rightarrow$ Classify finitely generated subgroups of $PL_+(S^1)$

INITIAL PROMPT

Give a rigorous solution to this mathematical research problem.

The solution should be at a publishable level.





SYLLABUS · COPE · AMS · LEIDEN DECLARATION (2026)

AI Use and Disclosure

Disclosure	Model, version, summary of prompts (not raw)
Novelty and credit	Literature search
Contribution	Who wrote which
Copyright	No unpublished work of others in commercial tools without permission

Agenda

1	**Class dates**	Lecture 2 and Lecture 3
2	**Five groups of five: discussion topics**	Email me three choices by Oct 20
3	**Discussion slides**	Email me by 1 day before Lecture 2
4	**Visualization OR Math Research**	Email me: **proposals** by Oct 20 **slides** by 1 day before presentation **notes** (1-3 pages) after the presentation